

Lab 02 Tasks

1. A food delivery service records customers, delivery riders, orders, and payments. Customers place orders using a mobile app. Riders use bikes provided by the company. The company has offices, rest areas, uniforms for riders, and CCTV cameras. Customers can rate riders after delivery.

Based on the above scenario, identify the following and provide reason/justification for each:

- Miniworld.
- Relevant entities & Irrelevant entities.
- Attributes and their data types.

2. A ride-sharing company operates in multiple cities. Users register using email and phone numbers. Drivers can use their own vehicles or company-provided vehicles. Each ride generates fare, time, distance, surge price, and payment method. Drivers receive ratings and bonuses. The company owns offices, parking lots, fuel stations, insurance policies, CCTV systems, and driver uniforms. The system also logs app crashes and user complaints.

Based on the above scenario, identify the following and provide reason/justification for each:

- List at least 6 relevant entities.
- List 5 things that should be excluded intentionally (A mix of entities and attributes).

3. An online bookstore sells both physical and e-books to customers across the country. The system keeps track of all books, including their title, author, genre, price, discount, stock quantity, publication year, edition, language, format, and ISBN. Customers can register with their name, email, and phone number, and place orders that may include multiple books. Each order records order date, delivery date, order status, payment method, and gift options. Customers can leave reviews with comments and ratings. The platform also tracks special promotions, seasonal discounts, bestseller tags, warehouse locations, and stock availability.

From the list below, choose only the relevant attributes for the “**Book**” entity:

- Book Title
- Author Name
- Genre
- Price
- Stock Quantity
- Discount
- Customer Name
- Order Date
- Review Comments
- Publication Year
- ISBN
- Format (hardcover, paperback, e-book)
- Language
- Warehouse Location
- Edition
- Payment Method

4. A fitness app tracks users, workouts, trainers, and diet plans. Each workout has name, type (cardio, strength, yoga), duration, calories burned, equipment required, and difficulty level. Diet plans include meal name, calories, ingredients, and serving size. Users have name, age, weight, height, subscription type, and activity logs. Trainers have name, certification, specialty, and session schedules. The app also tracks user devices, step counts, sleep patterns, water intake, and app crashes.

Based on the above scenario, if we pick “workout” as an entity, identify the following with reason/justification:

- At least 6 relevant attributes for the Workout entity
- Irrelevant info like user details, device logs, or trainer schedules should be excluded, explain why?